

SPEEDRUNBENCH: CHALLENGING LLM AGENTS WITH VIDEO GAME SPEEDRUNNING

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ABSTRACT

Frontier LLM agents have been shown to be capable of solving increasingly complex tasks for which humans have measurable solutions. This begs the pertinent question of whether LLM agents can go beyond what humans have already solved. The ability to develop sophisticated strategies to tackle consequential problems becomes paramount as well-trodden, human-developed solutions become insufficient for problems for which we lack context or enough training data. We study agents’ capability of such strategy formation through the communal practice of video game speedrunning. In speedrunning, practitioners compete to find the fastest way to complete a video game under certain conditions, and in so doing uncovering interesting unorthodox play styles that require a thorough understanding and mastery of the underlying game mechanics. We introduce **SPEEDRUNBENCH**, a benchmark that evaluates frontier LLM agents across 9 different games. To perform well in this benchmark, agents must repeatedly improve their strategy, reflect on their performance, exploit their gained knowledge, and reason across a long-horizon of actions to improve on an increasingly difficult problem: being faster than themselves and everyone else. Our experiments show that while frontier agents approach human world records in simple platformer games, they remain behind human performance on longer, more complex games under practical budgets. These results suggest that SPEEDRUNBENCH is a useful testbed for studying agents’ strategy formation capabilities as well as being a saturation-resistant evaluation measure, as there is almost always a faster completion time waiting to be discovered.¹

1 INTRODUCTION

Through the rich history of AI research, games have served as a common testbed for developing (Samuel, 1959; Tesauro, 1995), testing (Bellemare et al., 2013; Machado et al., 2018) and measuring (Mnih et al., 2015; Badia et al., 2020) various techniques, from search (Campbell et al., 2002; Silver et al., 2016) to state representation (Mnih et al., 2015; Hafner et al., 2023), and reinforcement learning (Silver et al., 2017; Vinyals et al., 2019; Berner et al., 2019). Games serve multiple desiderata for AI researchers: they are cheap, reproducible simulations with a clear ground truth (Bellemare et al., 2013; Kurach et al., 2020; Küttler et al., 2020), have human baselines (Mnih et al., 2015; Toromanoff et al., 2019), require multiple different skills across various modalities (Bellemare et al., 2013; Hafner, 2022; Reed et al., 2022), and many games require long-horizon reasoning under partial observability (Vinyals et al., 2017; Berner et al., 2019; Küttler et al., 2020). In being such a fruitful playground for AI development, games have been a driver of progress in their own right: developing AI to outplay humans in complex games has produced techniques that found uses elsewhere in AI research (Silver et al., 2018; Schrittwieser et al., 2020; Mandhane et al., 2022).

It is no surprise, then, that games are once again an epicenter of AI evaluation. Their desirable qualities still hold as the landscape has shifted towards LLM agents, and prior work has leveraged

¹Code: https://github.com/patronus-ai/speedrun_bench

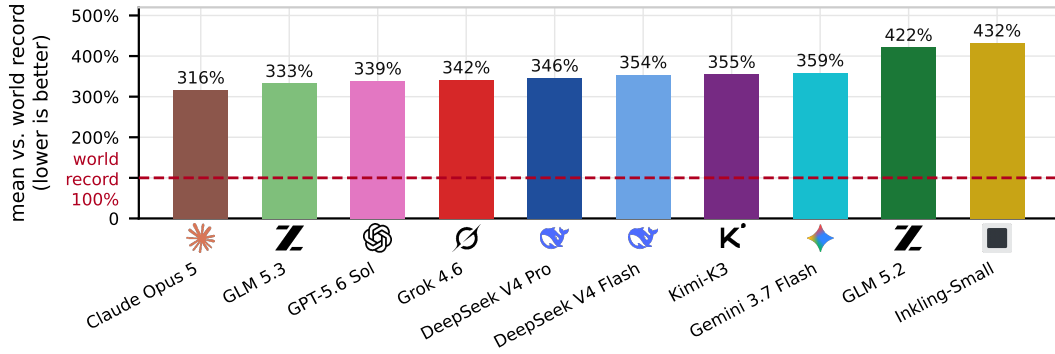


Figure 1: With the world record as an anchor (100%), we average the time the model took to reach the goal across the games and compare on how much longer it took than the world record as percentage (lower is better). Detailed game-level breakdown is in Figure 8.

many game genres (Côté et al., 2018; Hausknecht et al., 2020; Paglieri et al., 2025; Wu et al., 2024; Hu et al., 2026; Guertler et al., 2025; Zhang et al., 2025), and purpose-built game environments (ARC Prize Foundation, 2026) in assessing model performance. However, benchmarking for game-playing agents has so far focused on *completion*: progress toward, or attainment of, the end of the game (Paglieri et al., 2025; Zhang et al., 2025; Hu et al., 2026). While this remains a difficult task and, at the time of writing, a challenging one for frontier models (Zhang et al., 2025; ARC Prize Foundation, 2026), it does not provide researchers with an evergreen lens by which to continue to push and analyze model capabilities. No finite benchmark is immune to saturation, but some are more resistant, and we posit that *speedrunning* is a better fit for the current landscape of LLM agents evaluation. It is one thing for an agent to finish a complex game; it is another to be the fastest in the world to do so. Speedrunning is a community practice whereby players attempt to complete a game under specified criteria (e.g., with or without glitches) in the fastest possible time (Scully-Blaker, 2014). Players compete to beat the record, and discover strategies the game’s designers never anticipated (e.g., Figure 2). These strategies arise from the underlying dynamics of the game systems and speedrunners develop a more complete model of a game’s mechanics than the people that built it (Scully-Blaker, 2018). Game developers have responded in the past to speedrun records by patching their games, by deliberately preserving exploits, and by consulting speedrunners during development, with several studios reporting that speedrun analysis shaped design decisions (Scully-Blaker, 2018; Francis, 2016). Thus, speedrunning is a proven mechanism by which humans have developed a cooperative culture of strategy formulation to optimize a shared goal (finding the fastest possible route to the end) that had downstream consequences for other objectives.

As prior work has noted (Vinyals et al., 2019; Berner et al., 2019), games require long-horizon reasoning. Speedrunning stresses this further, as one must reason beyond what constitutes an acceptable solution to finding the fastest one. The state and action space of a game, even if held to be deterministic under a fixed seed, is finite but exponential in the machine’s memory (Hearn & Demaine, 2009), and it grows with every frame a run requires. If we take Super Mario Bros. for instance it has been shown to be NP-hard (Aloupis et al., 2015) to complete, and speedrunning requires an additional optimality constraint, which makes it a more complex problem (Krentel, 1988). Proving optimality in this context practically requires either a structural lower bound (e.g., the run must contain at least n unavoidable frames) or exhaustion of the reachable space. This the source of speedrunning’s resistance to saturation, a game can be observed to be unsaturated whenever a new run exceeds the known best time, but saturation itself cannot be proven without solving the optimization problem outright. Thus, in practical terms, we should expect LLM agents to find speedrunning substantially harder than completion, as the task does not saturate when the agent does a successful valid run, every new record is an upper bound upon which to improve, and this improvement requires counterfactual reasoning about alternative routes and mechanics that no single episode reveals. In the same way that beating the world’s experts at Go and Chess led to ground-breaking research in reinforcement learning (Silver et al., 2016; 2018), we believe speedrunning can invoke similar innovation in LLM agents.

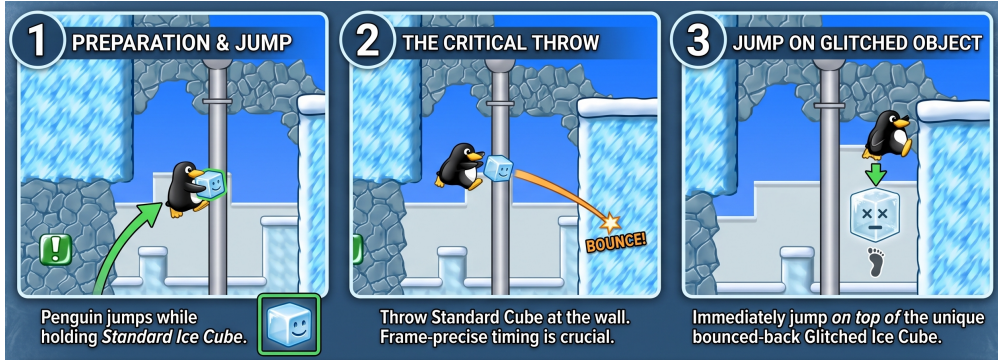


Figure 2: One example of a complex technique necessary for speedrunning SuperTux, an open-source 2D scroll action game. This technique is unnecessary to complete the game itself, but for speedrunning, this is critical to cut down the frame to reach the goal.

In this paper we introduce SPEEDRUNBENCH, a benchmark for evaluating LLM agents on speedrunning games, covering 10+ frontier agents and 9 different games. We evaluate agents in three settings: an ONLINE computer-use setting and two new offline settings. In ONLINE, an agent observes the current frame and selects the next action or short sequence of actions while the game runs. In both offline settings, the agent repeatedly edits and replays a complete frame-indexed action trace for a fixed number of iterations. OFFLINE-SEED begins with a goal-reaching reference trajectory and focuses on improving it, whereas OFFLINE-SCRATCH requires the agent to first construct a successful trajectory. Each setting highlights a different capability: ONLINE requires multimodal real-time perception, OFFLINE-SCRATCH requires route discovery, as the agent begins without a successful trajectory, and OFFLINE-SEED focuses on improving an existing solution, as the agent begins with a successful reference trajectory. All settings demand long-horizon coherence, since the agent has to keep improving a route it already holds, over many turns, without breaking it. Our experiments (Figure 1) show that while recent frontier agents approach human world records on simpler platformer games, they remain slower on long, complex games such as role-playing, racing, and strategy games, given a fixed budget. Even when given a successful trace as a starting point, agents achieve only marginal improvements, showing a gap between finding a valid solution and improving it.

2 RELATED WORK

Benchmarks for LLM Agents Seeing & Playing Games Games have been serving as evaluation benchmarks for machine learning (ML) models well before LLMs became popular, from text adventures (Côté et al., 2018) to 2D navigation (Guruprasad et al., 2026) and abstract reasoning tasks (ARC Prize Foundation, 2026). Games were also the setting for some of the field’s most visible results before LLMs, from Atari from pixels (Mnih et al., 2015) to superhuman play in Go (Silver et al., 2016) and StarCraft II (Vinyals et al., 2019). Recent benchmarks extend the setting to let frontier LLMs play games directly. LMGame-Bench (Hu et al., 2026) supplies perception and memory scaffolds, standardizes prompting, and screens for training data contamination, while VideoGameBench (Zhang et al., 2025) asks whether vision-language models (VLMs) can complete popular titles at all. Taesiri et al. (2024) pairs images with questions to evaluate whether a VLM can identify what is wrong in a rendered screen. Individual titles have also served as milestones in their own right, with Gemini 2.5 Pro clearing Pokémon² in 406.5 hours over two attempts (Comanici et al., 2025), and Hu et al. (2024) testing LLM capability on Pokémon battles specifically. Minecraft is the main setting where the objective is not merely completing something: MineDojo (Fan et al., 2022) and Voyager (Wang et al., 2024) score an agent on how much it acquires in an open world, by items obtained and skills learned. In contrast, SPEEDRUNBENCH scores how quickly it was completed, which continues to separate models after completion is reached by challenging them to learn and discover speedrun-related skills.

²Pokémon is a trademark of Nintendo Co., Ltd., Creatures Inc., and Game Freak Inc.

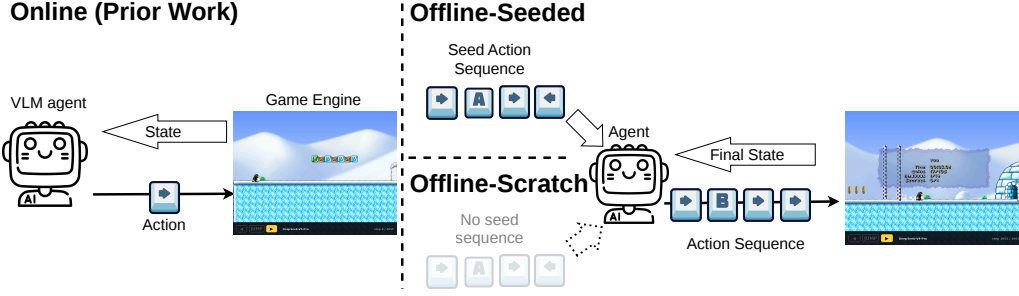


Figure 3: Overview of the different settings explored in SPEEDRUNBENCH: ONLINE, OFFLINE-SEED, and OFFLINE-SCRATCH. ONLINE setup predicts next action (or small action chunks) from a current state, whereas OFFLINE setup predicts the full action trace from the start to the goal.

Speedrunning as a Task Outside games, related work studies speedrunning LLM training (Zhao et al., 2025) and using agents to optimize LLM serving systems (Kamahori et al., 2026). The closest work to ours is the Pokeagent Challenge from NeurIPS 2025 (Karten et al., 2026a), in which speedrunning is one of the tracks, followed by Karten et al. (2026b) on online adaptation. Both evaluate an agent that observes a game frame and predicts the next action or a short chunk of actions, and both focus on a single title: Pokémon. SPEEDRUNBENCH differs in scope and in setting: our suite spans several genres, and beyond the online setting it evaluates an offline one in which the agent edits a complete frame-indexed trajectory, replays it, and scores on the resulting frame count, which turns the task from finding a working trajectory into improving one that already works.

3 SPEEDRUNNING WITH AN LLM AGENT

We first review what agent capabilities are required to perform well on speedrunning (§3.1), then introduce different setups for LLM agents speedrunning (§3.2). Using SPEEDRUNBENCH, we explore the following research questions:

- RQ1: Is SPEEDRUNBENCH challenging for modern LLM agents? Under what setup can a modern LLM-based agent speedrun, and under what setup can it not?
- RQ2: Can LLM agents optimize their gameplay with an autoresearch-like loop? Even for a near-optimal Tool-Assisted Speedrun (TAS) run?
- RQ3: What are the critical features for an LLM agent to speedrun?

3.1 AGENT CAPABILITIES MEASURED IN SPEEDRUNBENCH

Exploration: The agent needs to “think outside the box” to explore new paths and approaches to minimize the frames to the goal. This capability is tested especially in an OFFLINE-SCRATCH setup where given seed gameplay trajectories are not necessarily fast gameplays.

Discovery: For example, a glitch in Super Mario Brothers was discovered 40 years after the original game is released by a speedrunner (Colantonio, 2026).

Long-horizon Coherence: Traces run large number of actions, and a single wrong edit anywhere in one costs the whole run. The agent has to keep what already works while changing part of it, and to do so over hundreds of iterations.

3.2 ONLINE VS. OFFLINE SETTING FOR SPEEDRUNNING

We define a game as a deterministic environment with state space $\mathcal{S}$ and a finite action space $\mathcal{A}$, the set of controller inputs available on that console. Play begins from a fixed initial state $s_0 \in \mathcal{S}$, and each action advances the game by one frame, so a sequence of actions in $\mathcal{A}$ fully determines a run. We explore three settings per game: *Online*, *Offline-Scratch*, *Offline-Seed* for measuring different

Table 1: Games explored in SPEEDRUNBENCH, spanning several platforms and genres.

GAME	PLATFORM	GENRE
Super Mario Brothers	NES	Platformer
Super Mario Land	Game Boy	Platformer
Mario Kart 64	Nintendo 64	Racing
Pokémon Blue/Red	Game Boy	RPG
Civilization I	MS-DOS/SNES	Strategy
Astray	Browser	Puzzle
SuperTux	C++/Browser	Platformer
Tuxemon	Python/Browser	RPG
SuperTuxKart	C++/Browser	Racing

capabilities of each model (Figure 3). We further explain these three different setups in the following paragraphs.

Online (Action prediction from screenshots) Following prior work (Hu et al., 2026; Zhang et al., 2025), given a current state s of a game, the agent decides the next action $a \in \mathcal{A}$. A limited subset of frontier models (Opus 5, GPT-5.6-Sol, Grok 4.6) are capable of completing the game, but this setup only measures one aspect agent’s capability such as multimodal understanding. Since this setup can produce a baseline seed reference trajectory, we optimize it in the offline or trajectory editing setting.

Offline (Trace Editing) This setting mimics human speedrunning by playing the game over and over again throughout the decades. A run is represented as a frame-indexed action trace: a sequence of n actions $\{a_i\}_{i=1}^n \in \mathcal{A}$ covering an entire action sequence, from start to goal. On each turn the agent submits an edited trace, evaluated, and the agent receives the final state s_{final} of the trace such as 1) outcome $o \in \{\text{true}, \text{false}\}$ on whether it reached the goal, 2) the number of frames f to reach the goal if $o = \text{true}$, or the maximum progress the trace made if $o = \text{false}$ i.e., the edited trace no longer reached the goal. The agent repeats this for a fixed number of turns (200 unless specified). This can be regarded as action chunking i.e., predicting next k actions (Zhao et al., 2023) with an extremely long chunk. We explore the following two variant settings:

OFFLINE-SCRATCH: The agent has to come up with a valid action trave from scratch i.e., without a reference trace. This is harder than seeded setting due to a working trace not given in advance.

OFFLINE-SEED: The agent is given a seed reference trajectory to optimize from. Seed reference trajectory are either from humans, scripted by breadth-first-search (BFS), from online setup, which are all far from the best run for each game except for §5.2 where we analyze the effect of number of times calling the scorer.

Note that vision inputs are not required under OFFLINE settings, and therefore, more open-weight models can be evaluated under these settings. This is because, especially for simple games, no vision inputs are needed, smaller open-weight models can optimize.

3.3 AUTO-RESEARCH LOOP-BASED OPTIMIZATION

For OFFLINE setup, we use a self-improving loop inspired by auto-research (Karpathy, 2026) to mimic humans speedrunning by playing the game multiple times, see where time was lost, change one thing, and run it again, for as long as the route keeps giving. The OFFLINE settings reproduce that structure. At each iteration the agent selects its current best trace and the record of what it has already tried, proposes an edit, and receives a verdict. We fix both the number of iterations and how many candidate traces the agent may evaluate within one iteration before committing to the edit it submits. These hyperparameters are left fixed across models. (§5.1).

4 EXPERIMENTS

4.1 GAMES

Table 1 lists the nine titles in SPEEDRUNBENCH, spanning six genres and multiple consoles, and assembled along two axes. The first is genre, since the capability required for speedrunning depends on each game: a platformer requires frame-precise movement, an RPG rewards routing across a long horizon, and a puzzle game rewards search over a discrete state space. Alongside widely known commercial titles we include open-source games of the same genre, so that each of the three most popular and documented genres has a counterpart whose routes and records are far less likely to appear in a pretraining corpus.³

Super Mario Brothers, Super Mario Land, and SuperTux (The SuperTux Team, 2021) are 2D side-scrolling platformers, which demand precise movement and route discovery toward a level exit. The goal is finishing Level 1-1. Both Mario titles have decades of documented routes and world records, which supplies an external speedrun reference, and they are also the titles which are most likely to have been exposed to LLMs during pretraining. SuperTux is the open-source counterpart that makes the extent of exposure measurable: it has similar gameplay as Mario for the agent while carrying none of the same pretraining exposure. We use its “Welcome to Antarctica” level, referred to as SuperTux 1-1 throughout.

Pokémon Blue/Red and Tuxemon (Tuxemon Contributors, 2026) are turn-based RPGs, and the longest-horizon tasks in this benchmark. A run reaches the first gym badge only after 13 milestones, and the frame counts involved are one to two orders of magnitude larger than in the platformers, which stresses long-context tracking. Tuxemon is the open-source counterpart, matched in genre and structure but estimated to be less representative in the pretraining data in general.

Mario Kart 64 and SuperTuxKart (theodorepringle, 2021) are racing games, where the goal is lap time. We use Black Forest track for SuperTuxKart and Wario Stadium for Mario Kart 64.

Astray is a browser-based maze puzzle. We use the implementation from Ouyang et al. (2026).

Civilization I is a strategy game. These broaden genre coverage beyond the platformer and RPG settings that dominate prior game benchmarks.

4.2 SETUP

Models: We use frontier models both from proprietary models such as Claude Opus 5 (Anthropic, 2026), GPT-5.6-sol (OpenAI, 2026), Grok 4.6 (SpaceXAI, 2026), Gemini-3.7-Flash (Google DeepMind, 2026), and open-weight models such as Kimi-K3 (Team et al., 2026), GLM 5.3 (Z.ai, 2026b), GLM 5.2 (Z.ai, 2026a), DeepSeek-V4-Pro-0813 (DeepSeek-AI, 2026b), DeepSeek-V4-Flash-0731 (DeepSeek-AI, 2026a), Inkling-Small (Thinking Machines Lab, 2026). All models ran with OpenCode (OpenCode contributors, 2026) for experimenting on the same harness. Reasoning effort is set to the maximum available for every model in this paper.

Reference Times: Every world record we compare against is a human real-time attack (RTA) from verified runs (SINNED, 2025; EiP25, 2026; DylCat, 2025; theodorepringle, 2021; abney317, 2026).

Seed Reference Trajectory for OFFLINE-SEED: For traces to seed the agent in OFFLINE-SEED, GPT-5.6-sol/Opus 5 online run, scripted, and human traces are used. All the seed trajectories are far from the optimal speedrun play for leaving the space for agents optimize their play.

4.3 ONLINE PLAY RESULTS

Figure 4a shows ONLINE Pokémon play. Only the three vision-capable proprietary models reach the first badge at all, while Kimi K3 stalls at the prior point in the game where Reds(Ash) gets the Pokédex from Prof. Oak. All runs are still far off the from the human world record of $\sim 40,435$: Opus 5 takes 160,021 frames, GPT-5.6-sol 283,474, and Grok 4.6 477,140, so the best agent is roughly $4\times$ slower than the record and the worst nearly $12\times$. Under a completion metric, these three

³Commercial titles listed here are named just to identify evaluation setting. Users must obtain lawful copies themselves and nothing here grants rights beyond those already allowed by relevant licenses, TOS, & law.

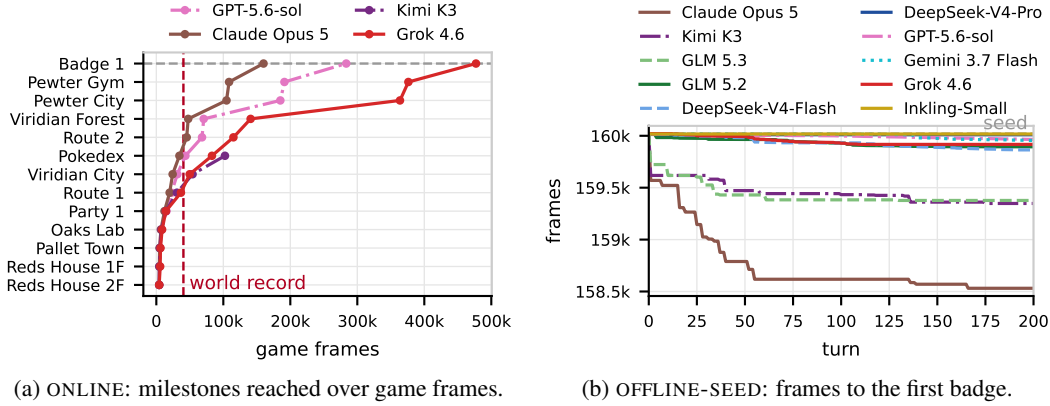


Figure 4: Pokemon Blue results. **(a)** ONLINE run on models reaching Pokemon Blue milestones. Opus 5 reaches Badge 1 in 160,021 frames, whereas the human world record is $\sim 40,435$ frames (SINNED, 2025): the record is $\sim 4\times$ faster than the best model result. Kimi K3 did not reach Route 2. **(b)** OFFLINE-SEED speedrun until the first badge, starting from the reference trajectory of Opus 5’s online run and given 200 iterations.

runs are indistinguishable, which shows the edge of SPEEDRUNBENCH at providing finer-grained discrimination.

4.4 OFFLINE SPEEDRUNNING RESULTS

As shown in Figure 1, no agent reaches a world record under the current budget, and the best mean result is more than twice slower than the world record. Figure 8 in Appendix A.2 gives the per-game breakdown. The closest results to the world record are the from-scratch SuperTux runs at 118%, while SuperTuxKart result are in between 256% and 261% and every model on Pokémon sits near 395%. A completion metric scores all of these runs identically.

In Table 2, every agent is run both from scratch and from a common 9498-frame seed produced by GPT-5.6-sol’s ONLINE run. All seven OFFLINE-SEED runs end within 12% of the seed, between 8,412 and 9,428 frames. In OFFLINE-SCRATCH, five of six models clear the level 3.8–4.9 $\times$ faster than the same model from the seed. GLM 5.2 is the exception: it reaches the goal only when handed the seed. OFFLINE-SCRATCH thus separates agents that can construct a route from those that can merely refine one, whereas OFFLINE-SEED shows how strongly a working but slow route through the game primes the agents.

On Super Mario Land 1-1, Claude Opus 5, GPT-5.6-sol, Grok 4.6 and GLM 5.3 cleared under OFFLINE-SCRATCH, at 2,835, 3,271, 3,417 and 4,113 frames, while the best OFFLINE-SEED run, from a 6,017-frame human seed, reached 5,145. On SuperTuxKart, under OFFLINE-SEED from a 168-second bot-generated lap, every model ends between 165 and 168 seconds. On Pokémon, again without the scorer, no OFFLINE-SEED run improves on the seed by more than 1% (Figure 4b). No model reaches the record on any of the three games.

Model	OFFLINE- SCRATCH	OFFLINE- SEED
<i>SuperTux 1-1</i>		
DeepSeek-V4-Pro	1,918	9,428
Gemini 3.7 Flash	1,951	9,094
DeepSeek-V4-Flash	1,980	9,123
GLM 5.3	2,087	8,412
Kimi K3	2,432	9,280
<i>Super Mario Land 1-1</i>		
Claude Opus 5	2,835	5,385
GPT-5.6-sol	3,271	5,896
Grok 4.6	3,417	5,730
GLM 5.3	4,113	5,534

Table 2: Frames to the goal under OFFLINE-SCRATCH and OFFLINE-SEED for the same model which found a trace to reach the goal (lower is better, with the faster in bold). SuperTux runs are seeded with a 9,498-frame trace, and Super Mario Land runs seed with a 6,017 trace. Results are largely influenced by the seed traces.

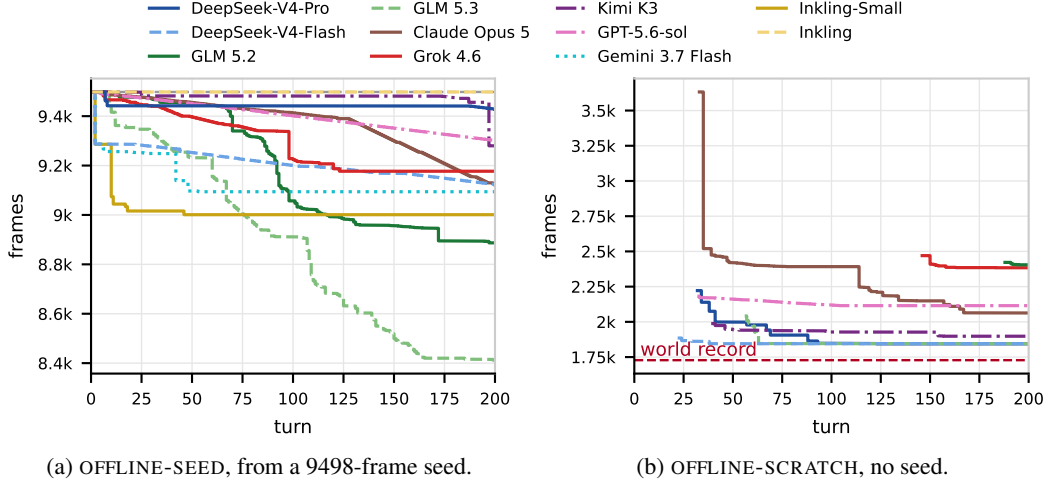


Figure 5: SuperTux 1-1 speedrun results with an agent backed with various LLMs. Lines in OFFLINE-SCRATCH start from the turn an agent finds a valid trace to complete the goal.

5 ANALYSIS

5.1 EXPANDING THE EVALUATION BUDGET PER TURN

In the experiments so far, the agent holds no in-turn scorer: it submits one trace per turn and gets one verdict. Here, we instead let the agent evaluate its own candidates, up to 200 per turn, and hold all else fixed. On SuperTux 1-1 this gives a controlled comparison, since the same models start from the same 9,498-frame seed and run for the same 200 turns in both settings (Table 3). With the scorer, every model escapes the seed and ends $4.0\text{--}5.1\times$ below where it ended without it. DeepSeek-V4-Pro reaches 1,847 frames, within 7% of the world record and about two seconds slower, while DeepSeek-V4-Flash and GLM 5.2 remain 16% and 29% above it (Figure 6a). On Super Mario Land 1-1, GLM 5.2 with the scorer comes within 3% of the record after 220K evaluated traces over ~ 114 hours (Figure 6b), against 5,643 frames from the same human seed without it. For these models, access to a scorer decides whether an agent escapes a non-optimal seeded trace, and given enough evaluations the strongest agent on each short 2D level comes close to its world record.

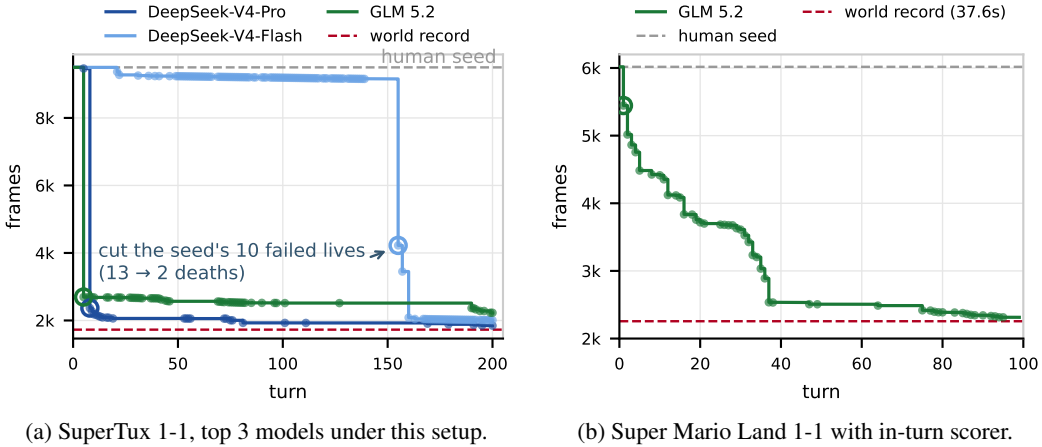


Figure 6: OFFLINE-SEED speedrun results with an agent allowed to evaluate up to 200 traces per turn. Super Mario Land world record is the 1-1 portion from EiP25 (2026). With this budget, GLM 5.2 on Super Mario Land & DeepSeek-V4-Pro on SuperTux come close to the world record.

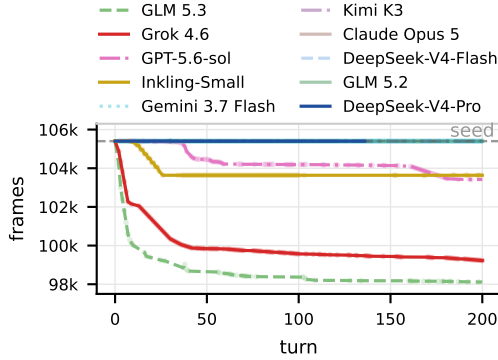


Figure 7: OFFLINE-SEED results on Tuxemon scored on frames to the first gym. All ten models start from the same 105,400-frame trace.

Model	no scorer	scorer
<i>SuperTux 1-1</i>		
DeepSeek-V4-Pro	9,428	1,847
Kimi-K3	9,280	1,930
DeepSeek-V4-Flash	9,123	2,009
GLM 5.2	8,887	2,228
<i>Super Mario Land 1-1</i>		
GLM 5.2	5,643	2,314

Table 3: Frames to the goal under OFFLINE-SEED for the same model and seed, without the in-turn scorer and with up to 200 evaluations per turn (lower is better).

5.2 CASE STUDY ON IMPROVING NEAR-OPTIMAL SPEEDRUN TRACE

The OFFLINE-SEED settings above start from seed traces far off the optimum, which leaves large room to improve for an agent. We report two experiments that instead start from a near-optimal trace on Super Mario Brothers: the first asks an agent to improve one which is same as OFFLINE-SEED. By seeding instead from an 18834-frame full-game Super Mario Brothers trace (Bisquit, 2003) and given 200 turns, GPT-5.6-sol returned a trace three frames faster. The second is merging two near-optimal traces⁴ which saved 26 frames from the fastest trajectory out of the two. These results suggest seeding the optimization loop with multiple near-optimal trajectory has the potential to further improve those.

5.3 CONTAMINATION FROM PRETRAINING: CASE STUDY ON POKEMON VS. TUXEMON

That famous titles are memorized in model parameters is well known: Hu et al. (2026) reports game content memorised during pretraining, and Gemini 2.5 Pro has been observed confusing items across Pokémon versions, expecting tea where the game requires soda (Zhang, 2025). Hence, we add a controlled pair: Pokémon and Tuxemon holding genre and setting fixed while varying how much a title is likely to appear in a pretraining corpus. Both are turn-based RPGs, run under OFFLINE-SEED with the same iteration budget and scored on frames to the first gym. Figure 7 shows that lower popularity does not make the open-source title harder to improve on when compared to the Pokémon run is the one in Figure 4b. No model improves on the Pokémon seed by more than 1%, whereas GLM 5.3 and Grok 4.6 improve on the Tuxemon seed by 6.9% and 5.9%. Tuxemon is harder to make any progress on since every model on Pokémon submits at least one trace faster than the seed, while four of ten never submit a trace that reaches the Tuxemon gym. Across ten models run on both titles, the rank correlation observed b/w titles is slightly negative at -0.18 . Hence, we see that results on a popular title are not predictive of performance on an unfamiliar one, which is consistent with popular titles being present in pretraining corpora.

6 CONCLUSION

We introduce SPEEDRUNBENCH, a benchmark that measures agents’ capability by assessing how fast an agent can finish a game and comparing it with the best known completion time. SPEEDRUNBENCH evaluates agents across three settings: an ONLINE setting where agents get frame by frame screen shots and must take an action, and two OFFLINE settings where agents must optimize their game plan for the run either with a seeded trace or without. We show that our benchmark remains challenging for current models (RQ1), as no agent reaches human world record on any game under a practical budget, with the best performing agent being on average *more than twice* as slow as the

⁴<https://tasvideos.org/UserFiles/Info/638242850875487965> and <https://tasvideos.org/UserFiles/Info/638586818800241580>

world record set by human players. Additionally we show that agents can improve their gameplay in an autoresearch-like loop (RQ2), even when the start seed is a near-optimal tool-assisted speedrun trace. However, giving the agents a slow seed as a starting point holds them back. Agents perform better when they can construct the routes of their own accord, and expanding the agents’ ability to evaluate a candidate with a larger per-turn budget, as seen with how close the platformer games came to the world records. We hope that SPEEDRUNBENCH inspires others to consider speedrunning as an avenue for pushing model abilities such as strategic thinking and long horizon coherence.

AI USE STATEMENT

We used generative AI tools to design or provide feedback on research methodology or experiments, to implement methods, to assist with translation, to clean and reformat datasets, and to support qualitative and thematic data analysis. We have not used generative AI tools to generate synthetic data sets, to help develop theoretical models or conceptual frameworks, to formulate mathematical claims, to provide critical ingredients for proving mathematical claims, to assist in the writing of proofs, or to propose or refine hypotheses. We have reviewed all AI-assisted work manually and take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

ETHICS STATEMENT

SPEEDRUNBENCH will be released for research and benchmarking purposes only and it will solely be limited to open-source games and licensed codes for public release to the community for reproducibility purpose. None of the copyrighted assets will be publicly released. This paper and associated assets conveys no right to access, reproduce, distribute, modify, or commercially exploit any third-party title or its assets beyond what applicable licenses, terms of service, and law already permit. Users are responsible for lawfully obtaining the titles they evaluate on and for securing any permissions their intended use requires.

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A APPENDIX

A.1 OPENSOURCE GAME CODE EXAMPLE

Our anonymized code for the patched version of Tuxemon is publicly available at <https://anonymous.4open.science/r/tuxemon-benchmark-anon-A2A4> under GPL-3.0 (code), CC-BY-SA (assets) following the original Tuxemon code and assets (Tuxemon Contributors, 2026).

A.2 PER-GAME RESULTS

Figure 8 shows the breakdown of results per game.

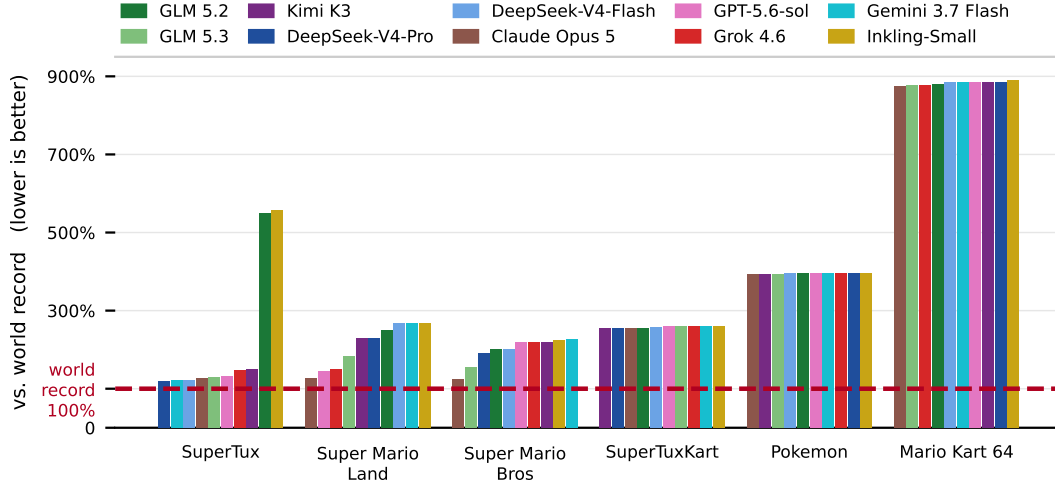
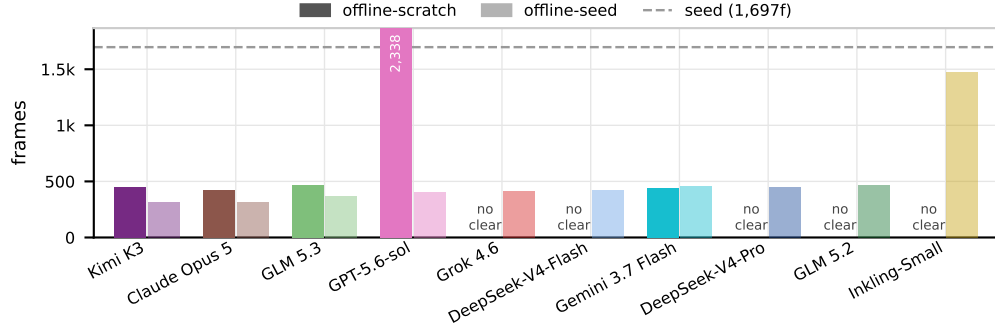
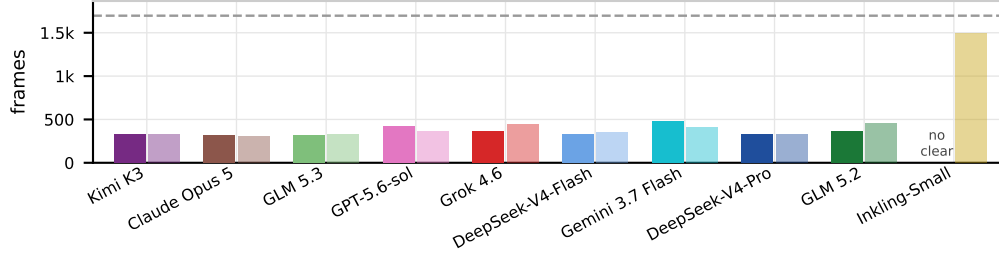


Figure 8: Per-game results: best result per model and title, as a percentage of the human world record (lower is better, and the dashed line is the record). Figure 1 averages every title here. Each bar is the better of that model’s OFFLINE-SCRATCH and OFFLINE-SEED run. Bars from runs that never beat the seed they were handed report the seed’s score rather than the model’s work. These are the Inkling-Small on SuperTux, the three right-most models in Super Mario Land, and the right-most models in SuperTuxKart. GLM 5.2 never cleared SuperTux from scratch, so its bar shows its seeded run. Mario Kart 64 has seeded runs only since no model cleared from scratch runs, and every model drives without using shortcuts, while the record uses a shortcut that skips most of the track.

Astray Figure 9 shows the Astray results.



(a) Game, browser, scorer and documentation all withheld.



(b) Scorer withheld, game and documentation accessible.

Figure 9: Final results on Astray level 1 under OFFLINE-SCRATCH (solid) and OFFLINE-SEED (lighter shade of the same colour), after 50 turns per run (lower is better, and the dashed line is the 1,697-frame seed). Both runs withhold the in-turn scorer. In (a) the agent also has no access to the game, a browser or the repository documentation. In (b) the game and documentation are available, so agents can load the maze and simulate it locally. Models are in the same order in both panels, ranked by (a), and each cell is a single run. GPT-5.6-sol’s OFFLINE-SCRATCH run in (a) reaches 2,338 frames and is clipped.

Civilization Across all models given 100 turns with no in-trun evaluations enabeled, none of the models improve the number of frames to reach the goal.

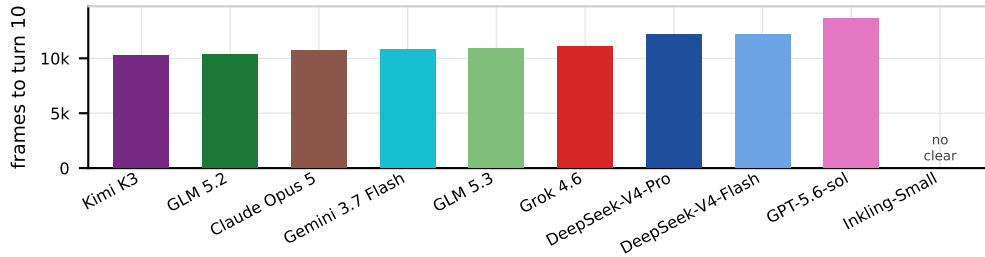


Figure 10: Frames to reach turn 10 of Civilization I, best verified run per model (lower is better). Every run had the in-turn scorer withheld and used its full 100 turns. Inking-Small never reached turn 10.